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Selection of Techniques and Metrics



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- Selecting An Evaluation Technique
- Selecting Performance Metrics
- **Commonly Used Performance Metrics**
- Utility Classification of Performance Metrics
- Setting Performance Requirements



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Commonly Used Performance Metrics

Outline of commonly used performance metrics:

- Response time
- Reaction time
- Throughput
- Capacity
- Reliability
- Availability

Response time

- the interval between a user's request and the system response,

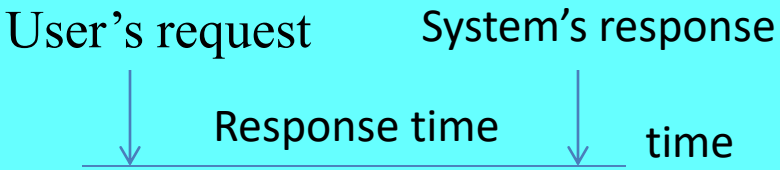


Fig: Instantaneous request and response

Two definitions:

- the interval between *the* ***end of a request*** *submission* and *the beginning of the corresponding response* from the system

- the interval between *the end of a request submission and the end of the corresponding response* from the system

Reaction time

- The time between submission of a request and the beginning of its execution by the system

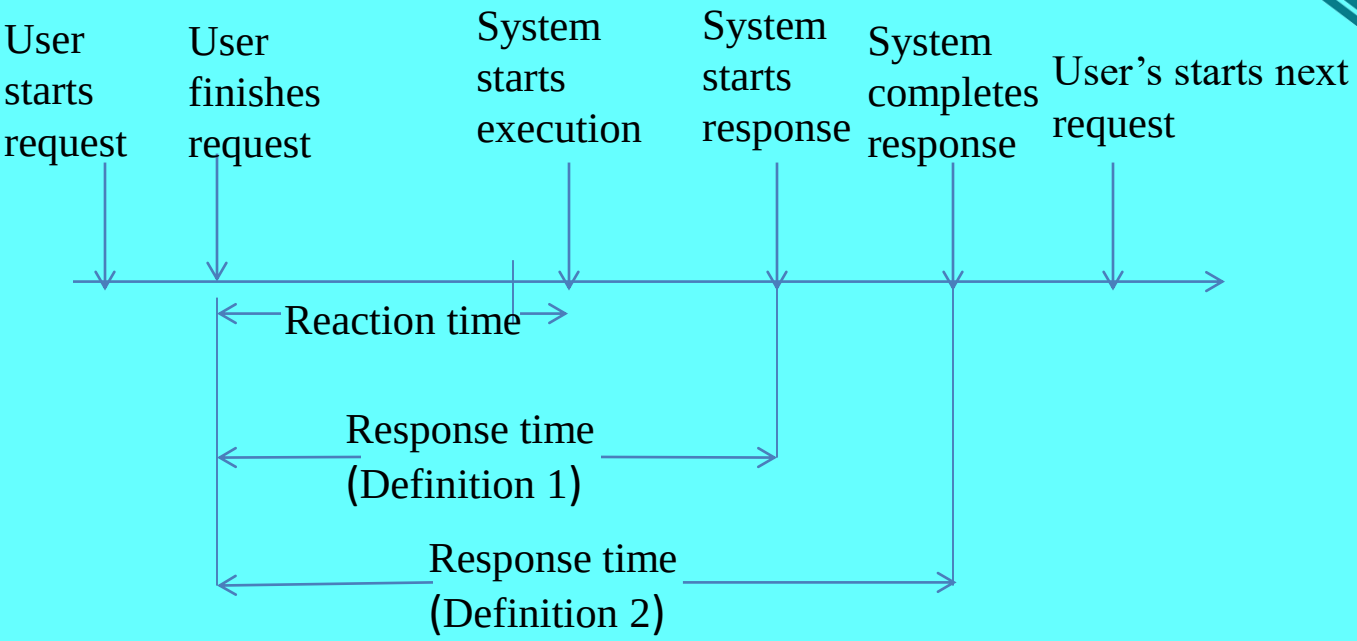


Fig : realistic request and response

Throughput

- the rate (requests per unit of time) at which the requests can be serviced by the system

For batch stream, jobs per second
For interactive systems, requests per second
For CPUs, Millions of Instructions Per Second **(MIPS)**, or Millions of Floating-Point Operations Per Second **(MFLOPS)**
For networks, packets per second **(pps)** or bits per second **(bps)**
For transactions processing systems, Transactions Per Second **(TPS)**

Capacity of a system

The maximum achievable throughput under ideal workload conditions is called **nominal capacity** of the system.
For computer networks,
the nominal capacity is called the **bandwidth** and is usually expressed in bits per second.

Usable capacity of the system

Response time at maximum throughput is more interesting to know the maximum throughput achievable without exceeding a pre-specified response time limit

The throughput at the knee is called the **knee capacity** of the system.

It is also common to measure capacity in terms of load, for example, the number of users rather than the throughput

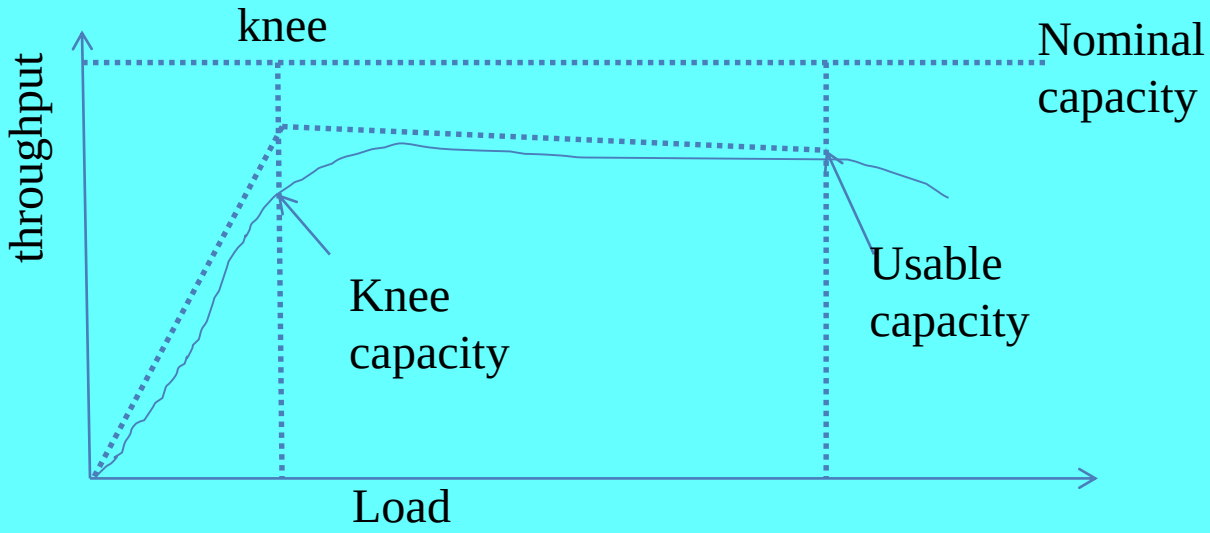


Fig : capacity of a system

Nominal Capacity: Maximum achievable throughput under ideal workload conditions.

E.g., bandwidth in bits per second. The response time at maximum throughput is too high.

Usable capacity: Maximum throughput achievable without exceeding a pre-specified response-time limit

Knee Capacity: Knee = Low response time and High throughput

- The **utilization** of a resource is measured as the fraction of time the resource is busy servicing requests

- The period during which a resource is not being used is called the **idle time**

- The **reliability** of a system is usually measured by the probability of errors or by the mean time between errors. The latter is often specified as **error-free seconds**

The **availability** of a system

- the fraction of the time the system is available to service user's requests
- The time during which the system is not available is called **downtime**
- the time during which the system is available is called **uptime**. Often the mean uptime, better known as the **Mean Time To Failure (MTTF)**

- The **cost/performance ratio** is commonly used as a metric for comparing two or more systems.

- The cost includes the cost of hardware/software licensing, installation, and maintenance over a given number of years

- For example, two transaction processing systems may be compared in terms of dollars per second (TPS).

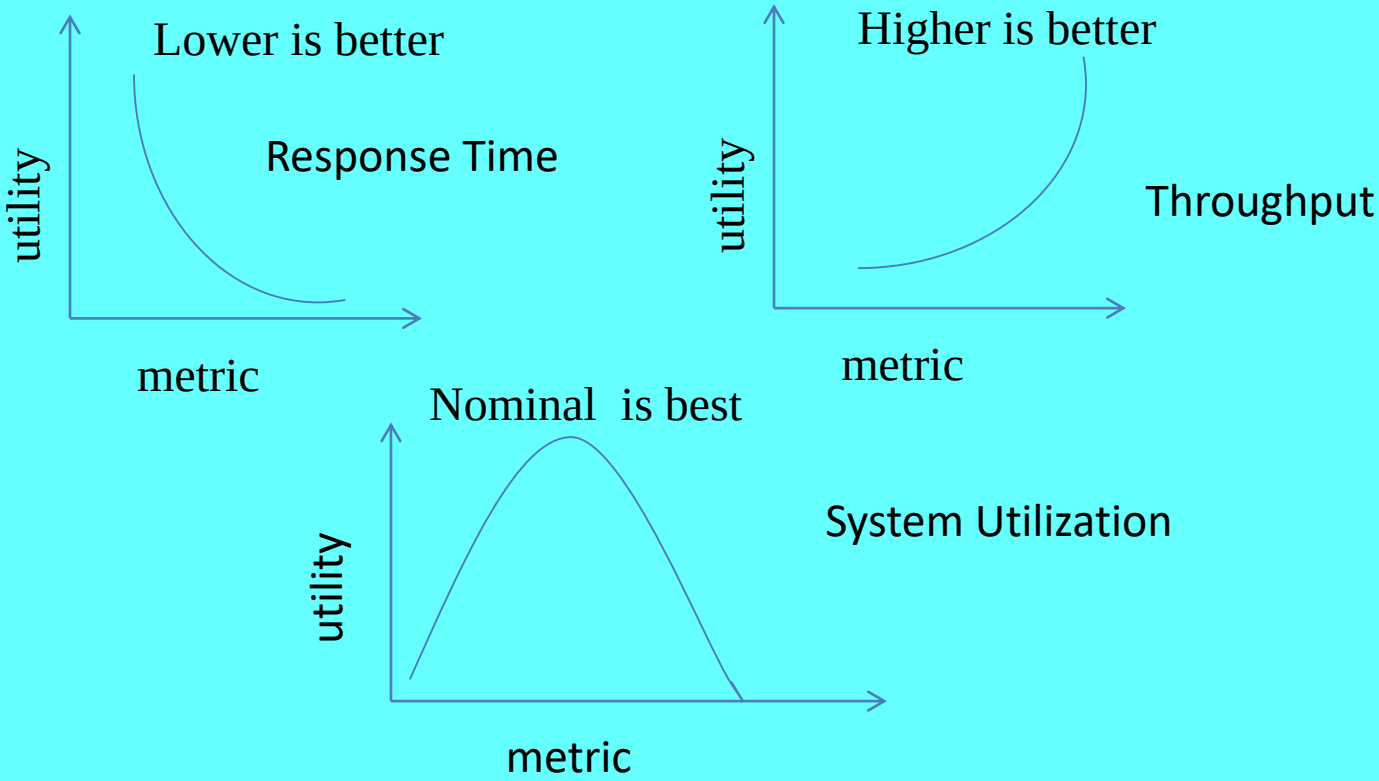
Utility Classification of Performance Metrics

Depending upon the utility function of a performance metric,

- Higher is Better or **HB**. System users and system managers prefer higher values of such metrics (e.g. *system throughput*)
- Lower is Better or **LB**. System users and system managers prefer smaller values of such metrics (e.g. *response time*)
- Nominal is Best or **NB**. Both high and low values are undesirable. A particular value in the middle is considered the best. (e.g. *utilization*)



Types of metric



Setting Performance Requirements

Unacceptable problems

1. Non-specific
2. Non-measurable
3. Non-acceptable
4. Non-realizable
5. Non-thorough



Acceptable problem

- SMART

Case Study: Two Congestion Control Algorithms

- Service: Send packets from specified source to specified destination in order.
- Possible outcomes:
  - Some packets are delivered in order to the correct destination.
  - Some packets are delivered out-of-order to the destination.
  - Some packets are delivered more than once (duplicates).
  - Some packets are dropped on the way (lost packets).

Case Study (Cont)

- Performance: For packets delivered in order,
  - Time-rate-resource ⇒
    - Response time to deliver the packets
    - Throughput: the number of packets per unit of time.
    - Processor time per packet on the source end system.
    - Processor time per packet on the destination end systems.
    - Processor time per packet on the intermediate systems.
  - Variability of the response time ⇒ Retransmissions
    - Response time: the delay inside the network

Case Study (Cont)

- Out-of-order packets consume buffers
  ⇒ Probability of out-of-order arrivals.
- Duplicate packets consume the network resources
  ⇒ Probability of duplicate packets
- Lost packets require retransmission
  ⇒ Probability of lost packets
- Too much loss cause disconnection
  ⇒ Probability of disconnect

EXERCISES

1. What methodology would you choose?

(a) To select a personal computer for yourself

(b) To select 1000 workstations for your company

(c) To compare two spread sheet packages

(d) To compare two data-flow architectures,

if the answer was required:

(i)Yesterday (ii) Next quarter (iii) Next year

2. Make a complete list of metrics to compare

(a) Two personal computers

(b) Two database systems

(c) Two disk drives and

(d) Two window systems


Thank you